

BTEC Level 3 Applied Science • Unit 1 Physics • Worksheet

Progressive Waves

Original JDSscience practice. These questions were written for this resource and are not official exam-board items.

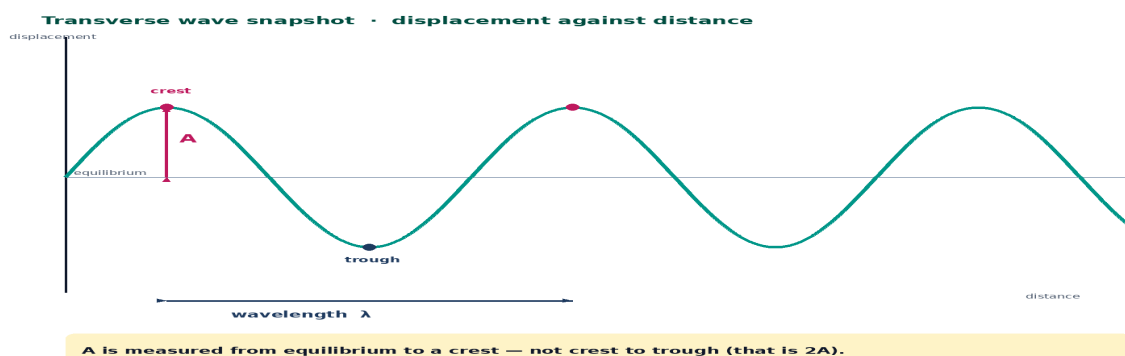
Learning focus: definitions, $v = f\lambda$, standard form and medium change.

A Retrieval

1. Define amplitude, wavelength and frequency.

2. Write the wave equation and give SI units for each quantity, including m s^{-1} .

3. How are period and frequency related?



4. On the snapshot, mark A and λ . How is amplitude read from a displacement–distance graph?

C Calculations

Use the same method every time: write the equation, convert to SI units, substitute, then give the unit.

Quantity	As written	SI value	SI unit

Use the blank rows to convert 200 MHz, 48 m / 0.16 s, and any other value before you substitute.

5. $f = 4.0 \text{ Hz}$, $\lambda = 0.80 \text{ m}$. Calculate v .

Working

6. A radio wave has $f = 200 \text{ MHz}$. Take $c = 3.00 \times 10^8 \text{ m s}^{-1}$. Calculate λ . Show the MHz $\rightarrow$ Hz conversion.

Working

7. A wave travels 48 m in 0.16 s. If $\lambda = 2.0 \text{ m}$, calculate f .

Working

D Exam-style practice

8. Define amplitude and wavelength. [2]

9. A note of 510 Hz travels at 340 m s^{-1} . Calculate λ . [2]

Working

10. Explain why air particles do not travel from a loudspeaker to a listener. [3]

E Challenge

11. A displacement–time trace has 5 complete cycles in 20 ms. The matching snapshot shows 4.0 cm between adjacent crests. Calculate f , T , λ and v . Show unit conversions.

Working

12. Which quantity stays the same when a wave enters a new medium? Explain what happens to v and λ .
